

GED Science Practice Test 3

PART A — QUESTIONS

GED-SCI3-001

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student investigates how temperature affects the activity of an enzyme that breaks down starch. Four test tubes contain the same amount of starch solution and enzyme solution. Each tube is kept at a different temperature for 10 minutes. The student then measures how much starch remains.

Temperature (°C): 10, 25, 37, 60

Starch remaining (mg): 18, 10, 3, 17

Which conclusion is best supported by the data?

- A) The enzyme worked best at 37°C.
- B) The enzyme worked best at 10°C.
- C) Temperature had no effect on enzyme activity.
- D) The enzyme created more starch at 60°C.

GED-SCI3-002

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student places four solid blocks into water. The density of water is 1.0 g/mL.

Block | Density (g/mL)

A | 0.6

B | 0.9

C | 1.4

D | 2.7

Which blocks should float in water?

- A) Blocks A and B
- B) Blocks B and C
- C) Blocks C and D
- D) Blocks A and D

GED-SCI3-003

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A geologist studies a cliff with several sedimentary rock layers. The lowest layer contains shell

fossils. A middle layer contains leaf fossils. The top layer contains small mammal fossils. The layers have not been folded or overturned.

Which statement is best supported by the information?

- A) The shell fossils are likely older than the mammal fossils.
- B) The top layer formed before the lowest layer.
- C) All fossils in the cliff are the same age.
- D) The leaf fossils formed after the mammal fossils.

GED-SCI3-004

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A person touches a hot pan and quickly pulls the hand away before thinking about the action.

Sensory neurons carry a signal from the skin to the spinal cord. Motor neurons then carry a signal from the spinal cord to muscles in the arm.

Which body system is mainly responsible for coordinating this rapid response?

- A) Digestive system
- B) Nervous system
- C) Respiratory system
- D) Reproductive system

GED-SCI3-005

Section: Science

Type: numeric_entry

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A cyclist travels 48 kilometers in 3 hours at a steady average speed.

What is the cyclist's average speed in kilometers per hour?

Type your answer as a whole number.

GED-SCI3-006

Section: Science

Type: multi_select

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Select TWO answers.

A student tests how the concentration of acid affects the rate at which magnesium reacts. The student places equal-length magnesium strips into acid solutions of different concentrations and measures the time until bubbling stops.

Which TWO factors should be kept constant to make the test fair?

- A) Length of each magnesium strip
- B) Temperature of each acid solution

- C) Concentration of each acid solution
- D) Time required for bubbling to stop
- E) Final reaction rate

GED-SCI3-007

Section: Science

Type: dropdown

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

In pea plants, the allele for tall stems is dominant over the allele for short stems. A plant with one tall allele and one short allele is [D1], and its visible trait is [D2].

Dropdowns:

D1) heterozygous | homozygous recessive | nonliving

D2) tall stems | short stems | no stems

GED-SCI3-008

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student measures the angle of sunlight hitting a surface at noon during different months.

Month | Sunlight angle at noon

March | 52°

June | 75°

September | 53°

December | 28°

Which month most likely has the greatest heating from direct sunlight?

A) March

B) June

C) September

D) December

GED-SCI3-009

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A sealed syringe contains air. When the plunger is pushed inward, the same amount of air occupies a smaller volume. The air pressure inside the syringe increases.

Which statement best explains why the pressure increases?

A) The air particles are destroyed.

B) The air particles collide with the syringe walls more often.

C) The air changes into a liquid immediately.

D) The syringe creates new air particles.

GED-SCI3-010

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A lake contains native plants, small fish, large fish, and birds. An invasive plant grows quickly across the lake surface and blocks sunlight from reaching underwater native plants.

Year | Native plant coverage (%) | Small fish counted

1 | 62 | 180

2 | 48 | 151

3 | 29 | 103

4 | 15 | 61

Which conclusion is best supported by the data?

- A) As native plant coverage decreased, the number of small fish also decreased.
- B) Small fish increased as native plant coverage decreased.
- C) Native plant coverage stayed constant for four years.
- D) The invasive plant improved small fish habitat every year.

GED-SCI3-011

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student wants to test which cup material keeps water warm the longest. The student pours 200 mL of hot water into cups made of foam, paper, glass, and metal. Each cup starts with water at 80°C. The student records the water temperature after 15 minutes.

What is the dependent variable?

- A) Type of cup material
- B) Starting water temperature
- C) Water temperature after 15 minutes
- D) Volume of water in each cup

GED-SCI3-012

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student measures the distance a rolling ball travels over time.

Time (s): 0, 1, 2, 3, 4

Distance (m): 0, 2, 4, 6, 8

Which statement best describes the ball's motion?

- A) The ball moves at a constant speed.
- B) The ball stops after 2 seconds.

- C) The ball slows down each second.
- D) The ball changes direction every second.

GED-SCI3-013

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

Freshwater moves across cell membranes by osmosis. If a plant cell is placed in very salty water, water moves out of the cell because the surrounding solution has a lower water concentration than the inside of the cell.

What is the most likely result?

- A) The plant cell loses water and shrinks.
- B) The plant cell produces extra chloroplasts immediately.
- C) The plant cell turns into an animal cell.
- D) The plant cell stops having a membrane.

GED-SCI3-014

Section: Science

Type: numeric_entry

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

A well draws 450 liters of groundwater in 5 hours.

What is the average pumping rate in liters per hour?

Type your answer as a whole number.

GED-SCI3-015

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student rubs a balloon on a wool sweater and then holds the balloon near small pieces of paper. The paper pieces jump toward the balloon.

Which interaction best explains this observation?

- A) Static electric charges attract the paper pieces.
- B) Gravity disappears near the balloon.
- C) The balloon changes the paper into metal.
- D) Sound waves pull the paper upward.

GED-SCI3-016

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

The kidneys help maintain homeostasis by filtering wastes from blood and adjusting the amount of water removed in urine. When a person is dehydrated, the body conserves more water, so less urine is produced.

Which statement best describes the role of the kidneys in this example?

- A) They help regulate water balance in the body.
- B) They pump oxygen-rich blood to muscles.
- C) They break down food in the stomach.
- D) They exchange CO₂ and O₂ in air sacs.

GED-SCI3-017

Section: Science

Type: dropdown

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

When ice melts on a sidewalk, solid water changes into [D1]. During melting, the ice [D2] thermal energy from its surroundings.

Dropdowns:

D1) liquid water | water vapor | carbon dioxide

D2) absorbs | produces | destroys

GED-SCI3-018

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A researcher tests whether a new bandage material helps small skin cuts heal faster. One group of volunteers uses the new bandage, while another similar group uses a standard bandage. The researcher records the number of days needed for each cut to close.

Why should the researcher include a group using a standard bandage?

- A) To provide a comparison for judging the new bandage
- B) To make the new bandage change its material
- C) To ensure that no cuts heal in either group
- D) To avoid collecting healing-time data

GED-SCI3-019

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student measures the brightness of a bulb when different numbers of identical batteries are connected correctly.

Number of batteries: 1, 2, 3

Bulb brightness rating: 2, 4, 6

What prediction is best supported if 4 batteries are connected in the same way?

- A) Brightness rating 0
- B) Brightness rating 2
- C) Brightness rating 8
- D) Brightness rating 12

GED-SCI3-020

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

A weather station records air pressure and wind speed as a storm approaches.

Time | Air pressure (mb) | Wind speed (km/h)

8 a.m. | 1008 | 20

10 a.m. | 1002 | 32

12 p.m. | 995 | 47

2 p.m. | 988 | 61

Which statement is best supported by the data?

- A) Wind speed increased as air pressure decreased.
- B) Wind speed decreased as air pressure decreased.
- C) Air pressure stayed constant throughout the day.
- D) The highest wind speed occurred at 8 a.m.

GED-SCI3-021

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A population of lizards lives on dark volcanic rock. Some lizards are dark, and some are light. Hawks can more easily see the light lizards against the dark rock. Over many generations, dark lizards become more common.

Which explanation best fits this change?

- A) Dark lizards were more likely to survive and reproduce in that environment.
- B) Light lizards changed color because they needed to hide.
- C) Hawks taught the lizards how to become darker.
- D) The environment stopped affecting survival.

GED-SCI3-022

Section: Science

Type: multi_select

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

Select TWO answers.

A class compares the energy available at different levels of a grassland food chain.

Level | Energy available (kJ)
Grasses | 10,000
Grasshoppers | 1,000
Frogs | 100
Snakes | 10

Which TWO statements are best supported by the table?

- A) Energy decreases at each higher level of the food chain.
- B) Grasses have the most available energy in this food chain.
- C) Snakes have more available energy than grasshoppers.
- D) Frogs produce more energy than grasses.
- E) Energy stays the same at all levels.

GED-SCI3-023

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student wants to know whether a rubber ball bounces higher on concrete or carpet. The student drops the same ball from 1 meter onto each surface and measures bounce height.

Which change would make the results more reliable?

- A) Repeat several drops on each surface and average the bounce heights.
- B) Use a different ball for each surface.
- C) Drop the ball from a different height each time.
- D) Measure only the first bounce on concrete.

GED-SCI3-024

Section: Science

Type: numeric_entry

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A sample of 200 seeds is tested. Of these seeds, 164 germinate.

What percent of the seeds germinate?

Type your answer as a percent rounded to the nearest whole number.

GED-SCI3-025

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A town uses several energy sources. Solar panels produce electricity only when sunlight is available. A coal power plant burns fuel that formed from ancient plant material over millions of years. Wind turbines generate electricity when moving air turns their blades.

Which energy source described is nonrenewable on a human time scale?

- A) Solar energy
- B) Coal
- C) Wind energy
- D) Moving air

GED-SCI3-026

Section: Science

Type: drag_drop

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

Match each target with the correct tile.

Tiles:

- A) Carries instructions inherited from parents
- B) Basic unit of structure and function in living things
- C) Group of similar cells working together
- D) Organ that pumps blood through the body

Targets:

T1) Cell

T2) Tissue

T3) DNA

GED-SCI3-027

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A factory compares the mass of iron and oxygen before and after forming rust in a closed container.

Mass before reaction:

Iron = 28 g

Oxygen = 12 g

Mass after reaction:

Rust = 40 g

Which statement is best supported by the data?

- A) Mass was conserved during the reaction.
- B) Oxygen had no mass before the reaction.
- C) The reaction destroyed 12 g of matter.
- D) Rust formed without using iron.

GED-SCI3-028

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A guitar string produces a higher-pitched sound when it is tightened. Tightening the string makes it vibrate faster when plucked.

What causes the higher pitch?

- A) A higher vibration frequency
- B) A lower vibration frequency
- C) A complete absence of waves
- D) A decrease in sound energy to zero

GED-SCI3-029

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A health worker records the average daily fiber intake of four adults.

Adult | Fiber intake (g/day)

- A | 11
- B | 18
- C | 27
- D | 22

Which adult has the highest daily fiber intake?

- A) Adult A
- B) Adult B
- C) Adult C
- D) Adult D

GED-SCI3-030

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student claims that smaller pieces of an effervescent tablet dissolve faster in water than a whole tablet. The student drops one whole tablet into 100 mL of water and drops one crushed tablet of the same mass into another 100 mL of water at the same temperature.

Which result would best support the claim?

- A) The crushed tablet stops fizzing sooner than the whole tablet.
- B) The whole tablet stops fizzing sooner than the crushed tablet.
- C) Both tablets stop fizzing at exactly the same time.
- D) The crushed tablet has a greater mass than the whole tablet.

GED-SCI3-031

Section: Science

Type: multi_select

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Select TWO answers.

A student investigates whether caffeine affects the heart rate of water fleas. The student measures heart rate before caffeine exposure and after exposure to a caffeine solution.

Which TWO choices would improve the investigation?

- A) Use the same caffeine concentration for all exposed water fleas.
- B) Measure several water fleas instead of only one.
- C) Change the microscope magnification during each count.
- D) Record only the result that matches the prediction.
- E) Use a different counting time for each water flea.

GED-SCI3-032

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

At a transform plate boundary, two plates slide past each other horizontally. Stress can build along the boundary until the rocks suddenly move.

Which event is most commonly associated with this type of plate boundary?

- A) Earthquake
- B) Lunar eclipse
- C) Hurricane
- D) River deposition

GED-SCI3-033

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A small population of island birds has two beak types: thin beaks and thick beaks. After a drought, most remaining seeds are large and hard. Scientists count the birds before and after the drought.

Beak type | Before drought | After drought

Thin beak | 64 | 22

Thick beak | 36 | 41

Which claim is best supported by the data?

- A) Thick-beaked birds likely had an advantage when large, hard seeds were common.
- B) Thin-beaked birds increased after the drought.
- C) Beak type had no relationship to survival during the drought.
- D) The drought caused every bird to have the same beak type.

GED-SCI3-034

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student compares how much work is done when moving different boxes. Work can be calculated as $\text{force} \times \text{distance}$.

Box	Force (N)	Distance moved (m)
A	10	3
B	15	2
C	20	4
D	25	1

Which box requires the greatest amount of work?

A) Box A

B) Box B

C) Box C

D) Box D

PART B — ANSWER KEY + EXPLANATIONS

GED-SCI3-001 — Correct: A — Correct Answer: The enzyme worked best at 37°C. —

Explanation: The least starch remained at 37°C, meaning the enzyme broke down the most starch at that temperature.

GED-SCI3-002 — Correct: A — Correct Answer: Blocks A and B — Explanation: Objects with density lower than water, 1.0 g/mL, float in water.

GED-SCI3-003 — Correct: A — Correct Answer: The shell fossils are likely older than the mammal fossils. — Explanation: In undisturbed sedimentary rock layers, lower layers are generally older than upper layers.

GED-SCI3-004 — Correct: B — Correct Answer: Nervous system — Explanation: The nervous system sends signals through sensory and motor neurons to coordinate rapid responses.

GED-SCI3-005 — Correct: 16 — Tolerance: exact — Correct Answer: 16 — Explanation: Average speed = distance $\div$ time = 48 km $\div$ 3 h = 16 km/h.

GED-SCI3-006 — Correct: A,B — Correct Answer: Length of each magnesium strip | Temperature of each acid solution — Explanation: These factors should stay the same so acid concentration is the main variable being tested.

GED-SCI3-007 — Correct: D1=heterozygous; D2=tall stems — Correct Answer:

D1=heterozygous; D2=tall stems — Explanation: One dominant and one recessive allele is heterozygous, and the dominant tall trait is visible.

GED-SCI3-008 — Correct: B — Correct Answer: June — Explanation: June has the highest listed sunlight angle, which means sunlight is most direct.

GED-SCI3-009 — Correct: B — Correct Answer: The air particles collide with the syringe walls more often. — Explanation: Compressing the gas puts the same number of particles in a smaller volume, increasing collision frequency and pressure.

GED-SCI3-010 — Correct: A — Correct Answer: As native plant coverage decreased, the number of small fish also decreased. — Explanation: The table shows both native plant coverage and small fish counts decreasing over time.

GED-SCI3-011 — Correct: C — Correct Answer: Water temperature after 15 minutes — Explanation: The dependent variable is what is measured in response to the cup material.

GED-SCI3-012 — Correct: A — Correct Answer: The ball moves at a constant speed. — Explanation: The ball travels 2 meters each second, so its speed is constant.

GED-SCI3-013 — Correct: A — Correct Answer: The plant cell loses water and shrinks. — Explanation: In salty water, water moves out of the plant cell by osmosis.

GED-SCI3-014 — Correct: 90 — Tolerance: exact — Correct Answer: 90 — Explanation: Average pumping rate = $450 \text{ liters} \div 5 \text{ hours} = 90 \text{ liters per hour}$.

GED-SCI3-015 — Correct: A — Correct Answer: Static electric charges attract the paper pieces. — Explanation: Rubbing the balloon creates static charge, which can attract small neutral or oppositely charged objects.

GED-SCI3-016 — Correct: A — Correct Answer: They help regulate water balance in the body. — Explanation: The kidneys adjust urine output to help maintain water balance.

GED-SCI3-017 — Correct: D1=liquid water; D2=absorbs — Correct Answer: D1=liquid water; D2=absorbs — Explanation: Melting changes solid ice into liquid water and requires the absorption of thermal energy.

GED-SCI3-018 — Correct: A — Correct Answer: To provide a comparison for judging the new bandage — Explanation: The standard-bandage group acts as a comparison group for evaluating the new bandage.

GED-SCI3-019 — Correct: C — Correct Answer: Brightness rating 8 — Explanation: The pattern increases by 2 brightness units for each additional battery, so 4 batteries would predict a rating of 8.

GED-SCI3-020 — Correct: A — Correct Answer: Wind speed increased as air pressure decreased. — Explanation: The table shows pressure decreasing from 1008 mb to 988 mb while wind speed increases from 20 km/h to 61 km/h.

GED-SCI3-021 — Correct: A — Correct Answer: Dark lizards were more likely to survive and reproduce in that environment. — Explanation: Dark coloring provided camouflage on dark rock, increasing survival and reproduction.

GED-SCI3-022 — Correct: A,B — Correct Answer: Energy decreases at each higher level of the food chain. | Grasses have the most available energy in this food chain. — Explanation: The table shows energy decreasing from producers to higher consumers, with grasses having the most available energy.

GED-SCI3-023 — Correct: A — Correct Answer: Repeat several drops on each surface and average the bounce heights. — Explanation: Repeated trials and averaging reduce the effect of random variation and improve reliability.

GED-SCI3-024 — Correct: 82 — Tolerance: exact — Correct Answer: 82 — Explanation: $164 \div 200 \times 100 = 82\%$, so 82% of the seeds germinated.

GED-SCI3-025 — Correct: B — Correct Answer: Coal — Explanation: Coal forms over millions of years and is nonrenewable on a human time scale.

GED-SCI3-026 — Correct: T1=B; T2=C; T3=A — Correct Answer: T1=Basic unit of structure and function in living things; T2=Group of similar cells working together; T3=Carries instructions inherited from parents — Explanation: Cells are the basic units of life, tissues are groups of similar cells, and DNA carries inherited instructions.

GED-SCI3-027 — Correct: A — Correct Answer: Mass was conserved during the reaction. — Explanation: The total starting mass was $28\text{ g} + 12\text{ g} = 40\text{ g}$, which equals the mass of rust produced.

GED-SCI3-028 — Correct: A — Correct Answer: A higher vibration frequency — Explanation: Higher-pitched sounds are produced by higher-frequency vibrations.

GED-SCI3-029 — Correct: C — Correct Answer: Adult C — Explanation: Adult C has the highest listed fiber intake at 27 g/day.

GED-SCI3-030 — Correct: A — Correct Answer: The crushed tablet stops fizzing sooner than the whole tablet. — Explanation: Crushing increases surface area, which can increase dissolving rate.

GED-SCI3-031 — Correct: A,B — Correct Answer: Use the same caffeine concentration for all exposed water fleas. | Measure several water fleas instead of only one. — Explanation: A consistent concentration controls the treatment, and testing several organisms improves reliability.

GED-SCI3-032 — Correct: A — Correct Answer: Earthquake — Explanation: Sudden movement along transform boundaries commonly produces earthquakes.

GED-SCI3-033 — Correct: A — Correct Answer: Thick-beaked birds likely had an advantage when large, hard seeds were common. — Explanation: Thick-beaked birds increased after the drought, while thin-beaked birds decreased.

GED-SCI3-034 — Correct: C — Correct Answer: Box C — Explanation: $\text{Work} = \text{force} \times \text{distance}$. Box C requires $20\text{ N} \times 4\text{ m} = 80\text{ J}$, the greatest value listed.